

Ivan SHYAKA

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☞ LinkedIn | GitHub

Personal Profile

Data Scientist and Software Engineer with strong capabilities in SQL, Python, and full-stack development (React, Node.js, Django, REST APIs), specializing in building data-driven applications and monitoring operational system performance. Currently completing a Master's in Big Data Analytics and experienced in developing dashboards, optimizing databases, and automating reporting pipelines that turn complex data into actionable insights. Skilled in DevOps practices notably containerization with Docker, version control with Git/GitHub, and cloud platforms (AWS, Azure) in order to build and maintain reliable, scalable systems. Additionally, experienced in building predictive models to forecast trends and drive data-backed decisions. Adept at collaborating with engineers and stakeholders to improve system reliability, operational efficiency, and data-informed decision-making.

Core Skills / Technologies

- **Programming:** Python, SQL, JavaScript
 - **Data Analysis:** Pandas, NumPy, Matplotlib, Data Cleaning, Exploratory Data Analysis, Operational Metrics Monitoring
 - **Data Visualization:** Power BI, Excel
 - **Databases:** PostgreSQL, MySQL, SQLite, NoSQL
 - **Development:** React, Node.js, Django, REST APIs, Nest.js, TypeScript
 - **Cloud / Platforms:** AWS, Azure, DHIS2
 - **Tools:** Git, GitHub
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Education

Master of Science in Big Data Analytics

Adventist University of Central Africa (AUCA) | December 2024 – Expected November 2026

Relevant Coursework: Advanced Database Management, Machine Learning for Big Data, Data Warehousing and Mining, Big Data Platforms, Predictive Modeling, and Data Visualization.

Bachelor of Science in Information Technology

Major: Software Engineering

Adventist University of Central Africa (AUCA) | March 2017 – November 2022

Relevant Coursework: Software Development Life Cycle (SDLC), Object-Oriented Programming (Java, C#), Database Systems, Web Development

(HTML/CSS/JavaScript), Software Testing & Quality Assurance, and Project Management.

Certifications & Training

Data Science Program

ALX – Africa | June 2025 – August 2026

- **Key Competencies:** Applied machine learning algorithms (regression, classification, natural language processing), statistical analysis, and data modeling to solve real-world problems.
- **Technical Skills:** proficient in Python programming (Pandas, NumPy, Matplotlib, Scikit-learn), SQL for database management and querying large datasets, and Power BI for data visualization and storytelling.

Data Analytics Course

ALX – Africa | October 2025 – January 2026

- **Key Competencies:** Expertise in gathering, cleaning, organizing, and interpreting large datasets to extract meaningful insights and trends.
- **Technical Skills:** Skilled in using spreadsheet software (Excel functions/formulas), SQL for data extraction and analysis, and data visualization with Power BI.

Professional Foundations Course

ALX - Africa | June 2025 – October 2025

- **Key Competencies:** Developed essential leadership and communication skills critical for success in the tech industry.
- **Professional Skills:** Strong capabilities in critical thinking, quantitative reasoning, entrepreneurial thinking, and managing complex tasks effectively.

React Development Course

Udemy | December 2024 – July 2025

- **Key Competencies:** Built modern, component-based web applications using React Hooks (useState, useEffect, useRef, useContext), state management with Redux, and client-side routing with React Router.
 - **Technical Skills:** Proficient in JSX, functional components, form handling and controlled inputs, HTTP requests/API integration, and TypeScript for type-safe React development; introductory exposure to Next.js for server-side rendering.
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Professional Experience

Data Systems & Operations Analyst (Hybrid) La Roca Group Ltd | January 2023 – Present

- Architected end-to-end commercial intelligence and billing solutions that synchronized real-time sales, procurement, and invoicing, ensuring 100% reconciliation accuracy and eliminating manual entry errors.
- Engineered automated reporting modules and dynamic dashboards to monitor high-velocity product trends and occupancy patterns, providing leadership with critical visibility into fiscal health and seasonal demand.
- Optimized inventory and hospitality operations by digitizing the reservation lifecycle and stock replenishment cycles, directly increasing room yield and resource efficiency through data-driven insights.
- Ensured high-level data integrity and system reliability by auditing complex databases, troubleshooting discrepancies, and translating technical datasets into actionable executive summaries and performance forecasts.
- Oversee the reliability and security of the custom-built system and broader IT environment, ensuring high uptime and data protection for all financial records.
- Led the transition to the new system by training staff across different functions, ensuring high adoption rates and minimal disruption to daily business activities.

Software Developer

Health Information System Provider (HISP) Rwanda | November 2024 – August 2025

- Developed and maintained web applications using different technologies such as React, Node.js, PostgreSQL and DHIS2.
- Optimized complex SQL queries and analyzed system data to diagnose performance issues and improve reliability of production health information systems.
- Debugged and troubleshooted software issues, ensuring smooth functionality.
- Collaborated with teams to gather requirements, plan development, and deploy solutions.
- Wrote clean, scalable, and well-documented code following best practices.
- Provided ongoing technical support and maintenance for web applications to ensure optimal performance.
- Stayed up-to-date with the latest industry trends and technologies, applying them to improve development processes.

Data Analyst & DevOps Officer

CYABINGO MINING COMPANY LTD (CMC) | January 2022 – October 2024

- Collected, cleaned, analyzed operational data from field activities to monitor trends, identify issues, and support decision-making.
- Produced regular reports and visual dashboards summarizing field performance, compliance gaps, and key operational indicators.
- Collaborated with field officers and local stakeholders to ensure accurate data collection and timely reporting from active sites.

- Supported incident tracking and resolution by analyzing case data, identifying patterns, and flagging anomalies in operational flows.
- Automated recurring report and dashboard pipelines to reduce manual processing time and ensure consistent, on-schedule delivery of operational data.
- Used Docker to containerize data processing and reporting tools, ensuring consistent environments across development and deployment.
- Managed version control and collaborative workflows using Git and GitHub, tracking changes to scripts and pipeline configurations.
- Maintained and monitored operational databases, overseeing data integrity, backups, and system uptime to support continuous reporting.

Data Analyst

Fabritech Ltd. | October 2020 – December 2021

- Monitored network performance metrics and service availability data, ensuring operational compliance and identifying system anomalies for timely resolution.
 - Collaborated with IT, Network Engineering, Customer Service, and Sales teams to deliver data-driven insights and support cross functional initiatives.
 - Analyzed subscriber growth trends, billing patterns, and customer behavior data to identify revenue opportunities and support business intelligence objectives.
 - Designed and maintained operational dashboards while ensuring data accuracy and consistency across systems for real-time business insights.
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References

Maurice Jules Mulisa

HISP Rwanda / Head of Software Development Department

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Kagenga Innocent

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Cyabingo Mining Company (CMC) / Founder & CEO

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